

Cheat Sheet: Vector Search with NoSQL Databases

Term/Function/Module	Description	Example
Docker exec	Command used to run a specific command within a running Docker container.	<code>docker exec -it {container_id} bash</code>
Docker ps	Command to list currently running Docker containers.	<code>docker ps -a</code>
Docker run	A command used to create a container from a Docker image and start the container.	<code>docker run --name cassandra -d -p 9042:9042 cassandra:5.0</code>
dotenv	A module that loads environment variables from a <code>.env</code> file into <code>process.env</code> in Node.js projects.	<pre>require('dotenv').config(); console.log(`\${process.env.CASSANDRA_KEYSPACE}`);</pre>
dotproduct Similarity	Measures the similarity between two vectors and used in machine learning to determine the angle between vectors.	
Embeddings	Dense representations of text or other types of data, used in machine learning to capture the semantic	

	properties of the data.	
embeddingsRequest.arraySync	A method in TensorFlow.js that synchronously returns the result of a tensor operation, useful in scenarios where the tensor result is needed immediately.	<pre>const vector = embeddingsRequest.arraySync()[0];</pre>
Euclidean Similarity	Often referred to as Euclidean distance; a common measure of distance between two points in Euclidean space, used in various machine learning algorithms.	
export	A keyword in JavaScript used to export functions, objects, or primitive values from one module so they can be used by other programs.	<pre>module.exports = { vectorizeText, connectToMongoDB, closeMongoDBConnection, MONGO_COLLECTION };</pre>
Express	A web application framework for Node.js, designed for building web applications and APIs with a minimal setup.	<pre>npm install express const app = express(); app.use(express.json());</pre>
for of loop	A modern JavaScript loop that creates a loop iterating over iterable objects, simplifying syntax compared to traditional for loops.	<pre>for (const newsItem of newsArray) { ... }</pre>
Image	In Docker, a static file that is	<pre>docker images</pre>

	used to produce a container, typically containing all necessary executable code to run an app.	
Indexing	The process of creating a data structure that improves the speed of data retrieval operations on a database.	<pre>const indexQuery = `CREATE INDEX IF NOT EXISTS ann_index on \${process.env.CASSANDRA_KEYSPACE}.\${process.env.CASSANDRA_TABLE} (embedding) USING 'sai';`;</pre>
Kaggle	An online community and platform for predictive modeling and analytics competitions and data sets.	
Model	In machine learning, a model represents what has been learned by a training algorithm; it takes input data and makes predictions or decisions based on the data.	
model.embed	A method used in TensorFlow, to convert raw data such as text into a numerical array of fixed size, called an embedding.	<pre>const embeddings = await model.embed([text]);</pre>
MongoDB	A NoSQL database known for its high performance, high availability, and easy scalability.	
mongodb	Node.js package to interact with MongoDB.	<pre>npm install mongodb</pre>

MongoDB Atlas	A service by MongoDB designed to host MongoDB databases in a fully managed, cloud-based environment.	
mongosh	The MongoDB Shell, `mongosh`, is a fully functional JavaScript and Node.js environment for interacting with MongoDB deployments.	<pre> mongosh --host localhost --port 27017 --username root --password root </pre>
npm	The default package manager for the JavaScript runtime environment Node.js.	
npm init	A command-line tool to create a new Node.js project, initializing a `package.json` file with the project details.	<pre> npm init -y </pre>
npm install	A command used in Node.js to install dependencies listed in `package.json`.	<pre> npm install npm install express </pre>

replication_factor	In Cassandra, a setting that determines the number of copies of data that will be stored across different nodes to ensure fault tolerance.	<pre>CREATE KEYSPACE IF NOT EXISTS \${process.env.CASSANDRA_KEYSPACE} WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};</pre>
require	A built-in function in Node.js used to include modules in your application, particularly for importing functionalities from libraries.	<pre>require('dotenv').config();</pre>
res.status	A method in Express.js used to set the HTTP status for the response.	<pre>return res.status(400).send({ error: 'Text is required' });</pre>
res.status.send	A chainable method in Express.js that sets the HTTP status code and sends a text response back to the client.	<pre>return res.status(400).send({ error: 'Text is required' });</pre>
REST	Representational State Transfer, an architectural style for distributed hypermedia systems like the World Wide Web.	
sai	Typically refers to Storage Attached Indexing in Apache Cassandra, allowing more efficient data indexing and querying.	
Similarity Function	A function that computes the	

	degree of similarity between two data points, often used in data mining and machine learning.	
TensorFlow	An open-source framework developed by Google to conduct machine learning and deep neural network research.	
touch	A command used to create a new, empty file or to update the access and modification times of an existing file.	<code>touch filename.js</code>
use myvectordb	A command in MongoDB to switch the context to a specific database called 'myvectordb'.	<code>use myvectordb;</code>
VECTOR <FLOAT, 512>	Specifies a data type in some database systems, indicating a vector of floating point numbers with 512 dimensions.	
Vectors	Arrays of numbers that represent data in multiple dimensions, commonly used in machine learning to represent features.	
wget	A command-line utility used for downloading files from the web, supporting HTTP, HTTPS,	<code>wget -O \$VECTOR_HOME/data/songs.csv {URL}</code>

	and FTP protocols.	
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